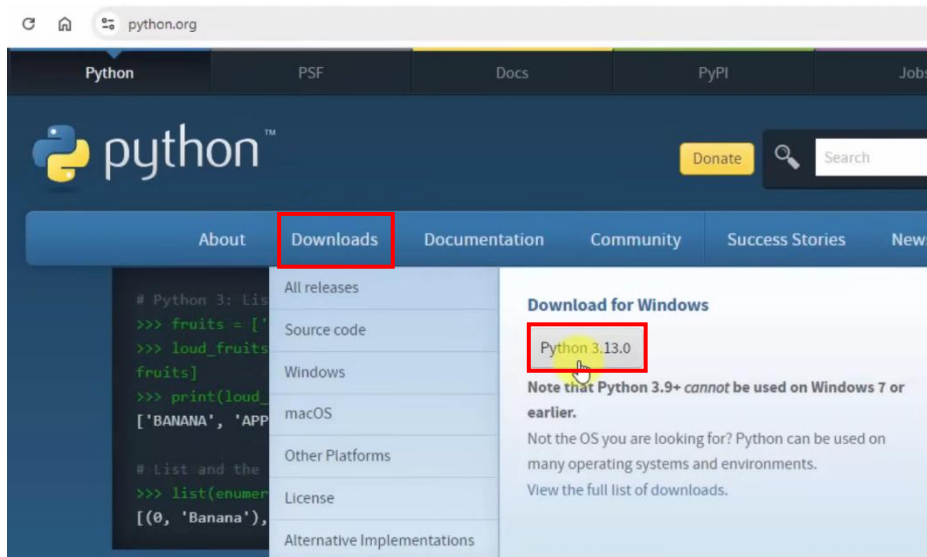


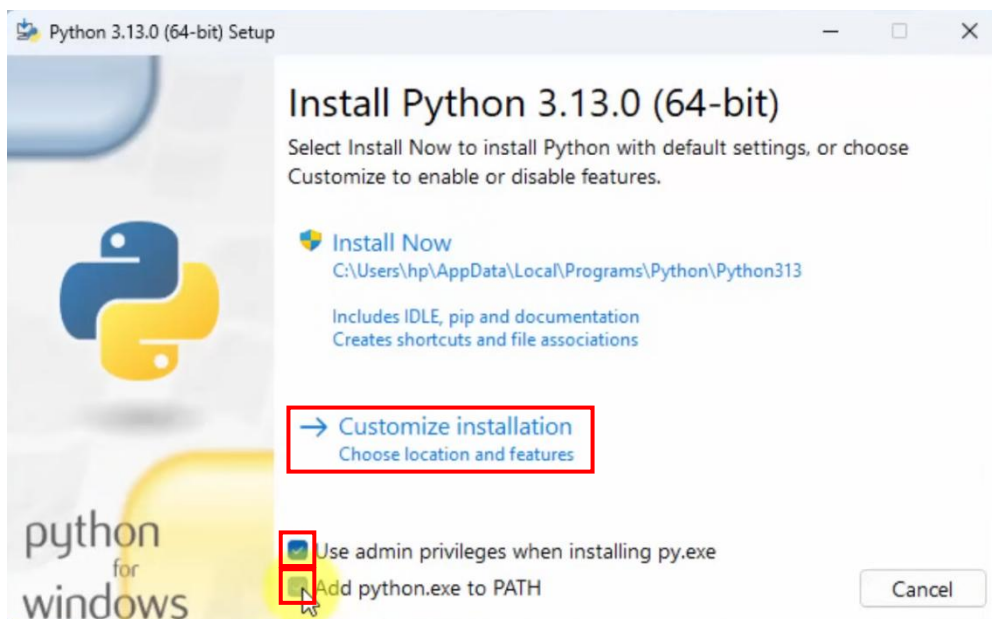
Install Python environment on your Windows

1. Click Downloads > Python 3.x.x

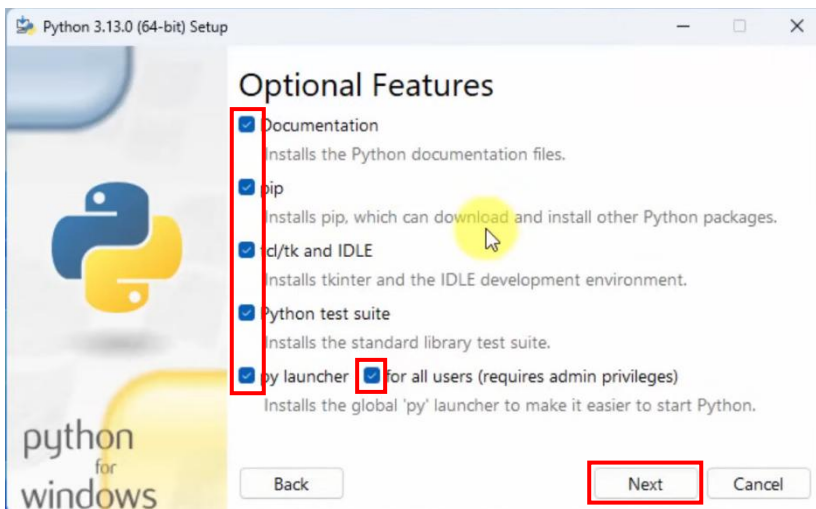


2. Check:

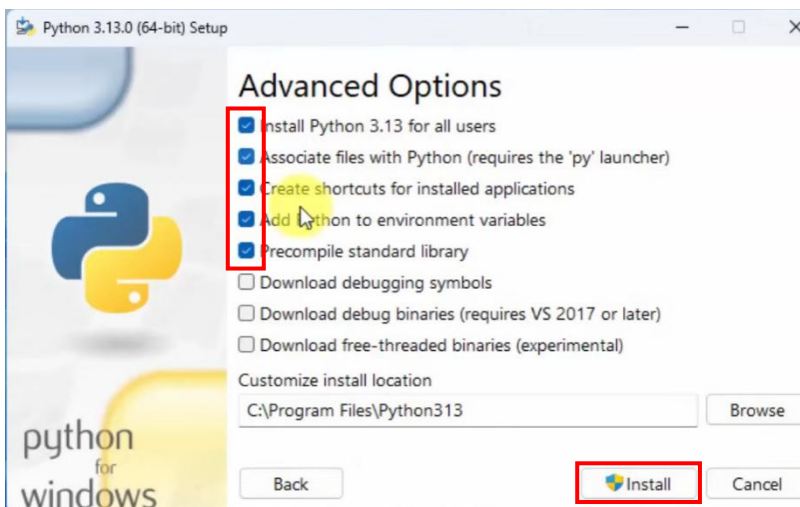
- Use admin privileges when installing py.exe
- Add python.exe to PATH
- > Customize installation



3. Check all > Next



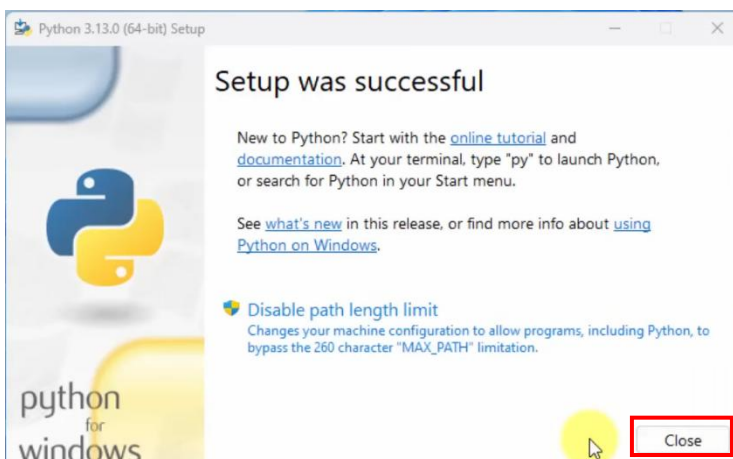
4. Advanced options



Check:

- Install Python 3.13 for all users
- Associate files with Python
- Create shortcuts for installed applications
- Add Python to environment variables
- Precompile standard library
- > Install

5. Complete installation: click Close



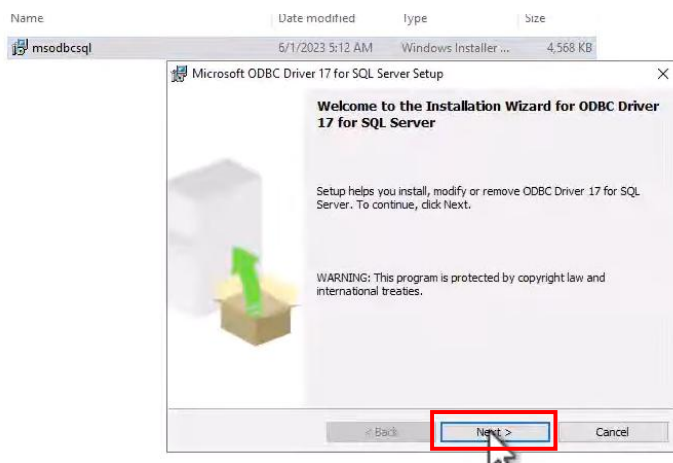
ODBC SQL Server Connection

6. Install ODBC Driver17

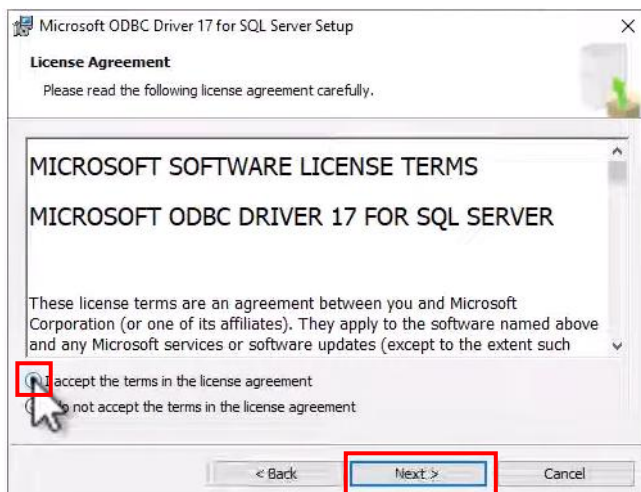
Url: <https://learn.microsoft.com/en-us/sql/connect/odbc/download-odbc-driver-for-sql-server?view=sql-server-ver17>

The screenshot shows the Microsoft ODBC Driver 17 for SQL Server download page. The page title is "Microsoft ODBC Driver 18.6.2.1 for SQL Server (ARM64): Chinese (Simplified) | Chinese (Traditional) | Czech | English (United States) | French | German | Italian | Japanese | Korean | Polish | Portuguese (Brazil) | Russian | Spanish | Turkish". The page is in English. The "Version 17" section states: "Version 17.11.1 is the latest general availability (GA) version of the 17.x driver. If you have a previous version of Microsoft ODBC Driver 17 for SQL Server installed, installing 17.11.1 upgrades it to 17.11.1." Below this, there are two download links: "Download Microsoft ODBC Driver 17 for SQL Server (x64)" and "Download Microsoft ODBC Driver 17 for SQL Server (x86)". A list of release information is provided: "Release number: 17.11.1.1" and "Released: April 30, 2026". At the bottom, it says: "This release of Microsoft ODBC Driver for SQL Server can be installed in the following languages:".

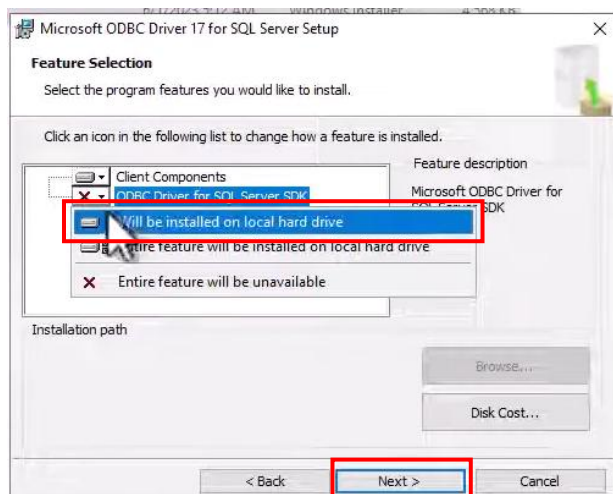
Click next



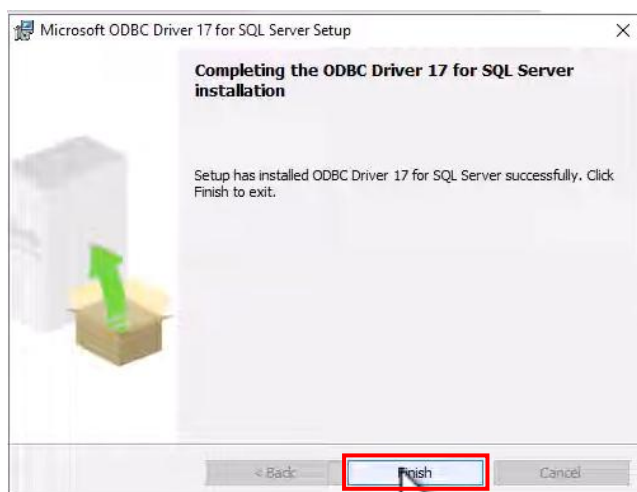
Check: I accept the term in the license agreement > Next



Click ODBC Driver for SQL Server SDK > Will be installed on local hard drive > Next

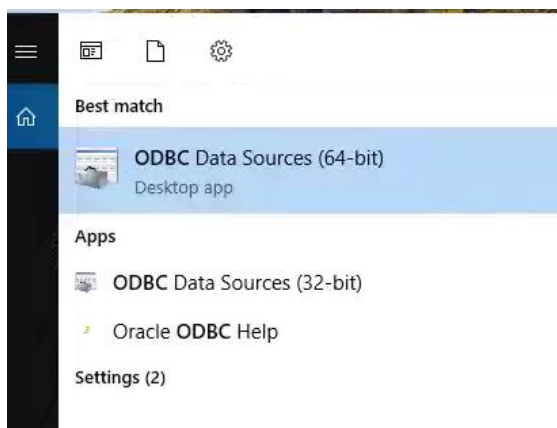


Next > Finish

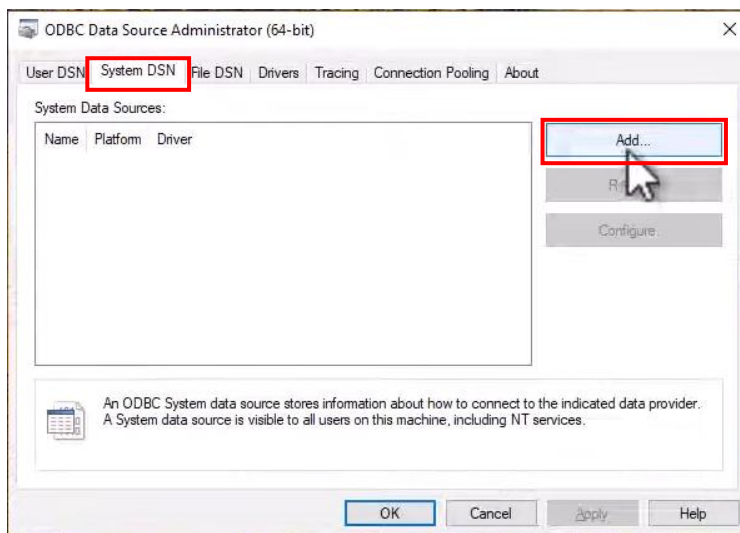


Configure ODBC

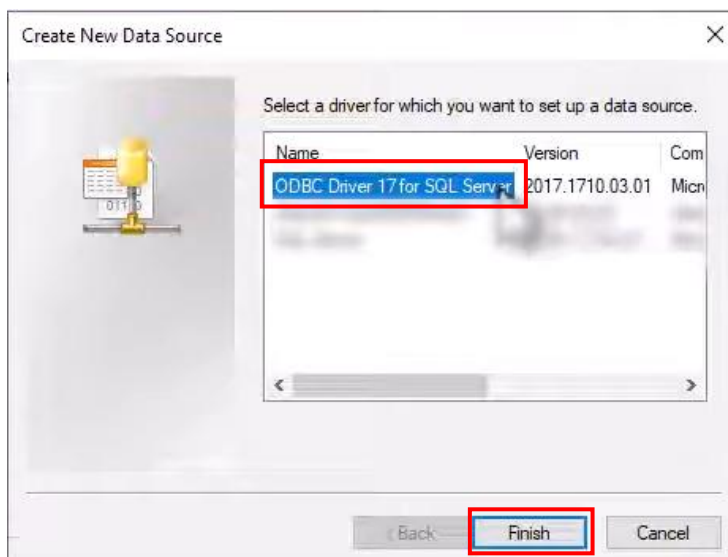
7. Search ODBC Data Sources (64-bit)



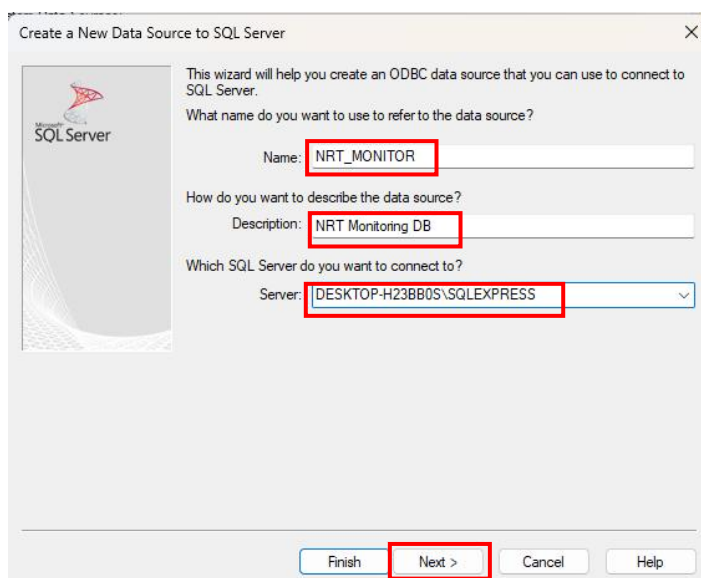
Click System DSN > Add



Choose ODBC Driver 17 for SQL Server



Fill in the details accordingly



Name: NRT_MONITOR
Description: NRT Monitoring DB
Server: <Choose your ssms server name>
> Next

Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

☐ With Integrated Windows authentication.

SPN [Optional]:

☐ With Azure Active Directory Integrated authentication.

☒ With SQL Server authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.

☐ With Azure Managed Service Identity authentication.

☐ With Azure Service Principal authentication.

Login ID: sa

Password:

< Back Next > Cancel Help

If your server is unauthenticated, choose “With Integrated Windows authentication” > Next.

Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

☒ With Integrated Windows authentication.

SPN (Optional):

☐ With Azure Active Directory Integrated authentication.

☐ With SQL Server authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.

☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.

☐ With Azure Managed Service Identity authentication.

☐ With Azure Service Principal authentication.

Login ID:

Password:

< Back Next > Cancel Help

Create a New Data Source to SQL Server

☒ Change the default database to:

master
model
msdb
tempdb
ParkingServerDB

☐ Attach database filename:

☒ Use ANSI quoted identifiers.

☒ Use ANSI nulls, paddings and warnings.

Application intent:
READWRITE

☐ Multi-subnet failover.

☒ Transparent Network IP Resolution.

☐ Column Encryption.

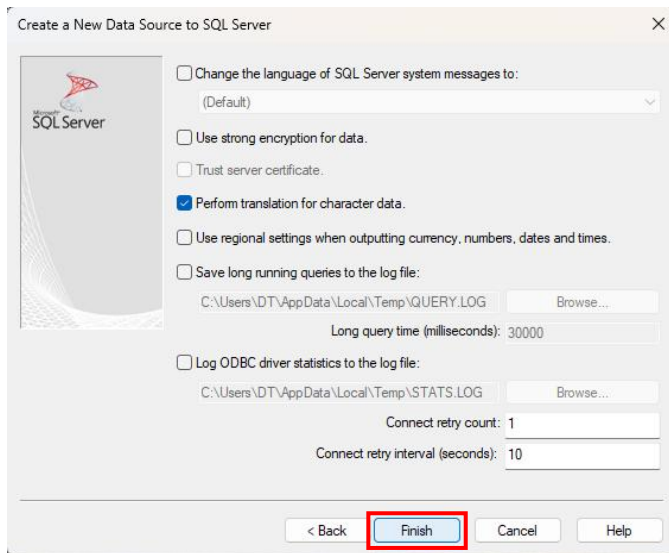
Enclave Attestation Info:

Keystore Configuration...

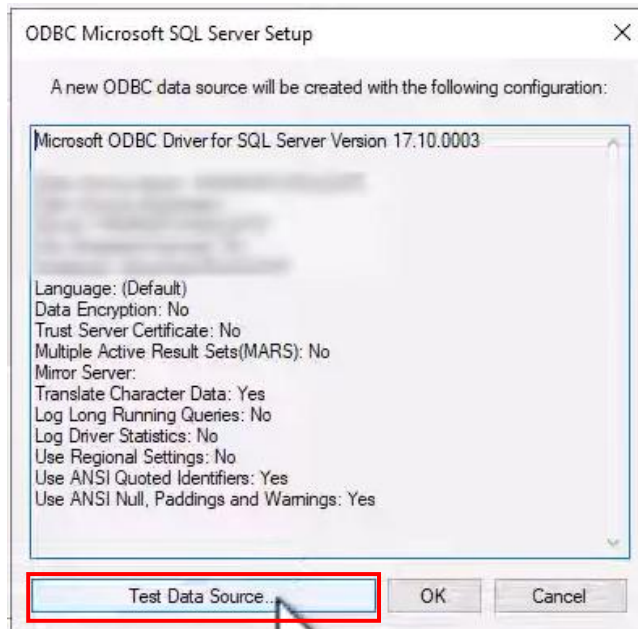
☐ Use FIMONLY metadata discovery.

< Back **Next >** Cancel Help

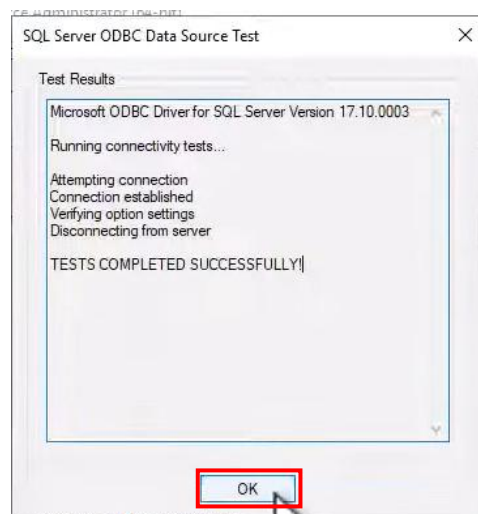
Click Finish



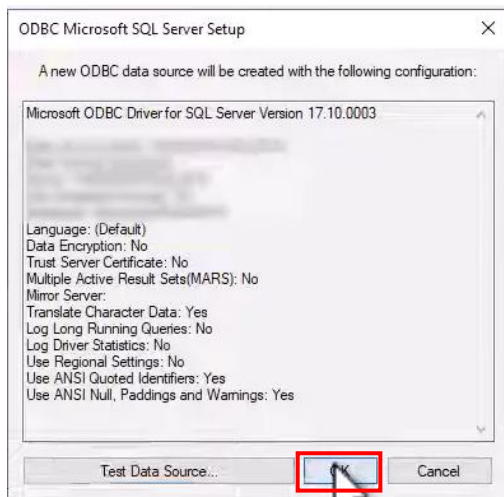
Click Test Data Source



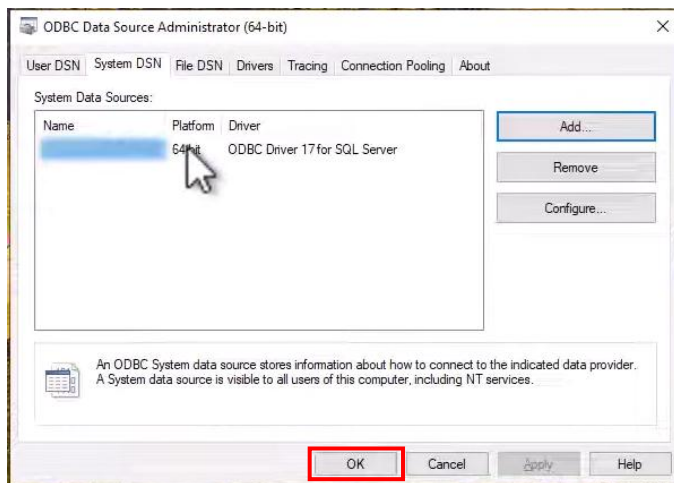
If you see 'Connection established', it means the connection is successful > click OK



Click OK



Click OK



Run Instruction

1. **Change directory**

```
cd NRT_Monitor
```

2. **Create a virtual environment**

```
py -m venv venv  
.\venv\Scripts\activate # On Linux Mac OS: source venv/bin/activate
```

- If errors occurred, run this:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

3. **Install dependencies**

```
pip install -r requirements.txt
```

4. Credentials configuration

In the Agent.py

```
SQL_SERVER = r"DESKTOP-H23BB0S\SQLEXPRESS"  
DATABASE = "NRT-V2-Test"  
SITE_NAME = "Test Site"  
SITE_CODE = DATABASE  
API_URL = "https://script.google.com/macros/s/AKfycbwwnMbJ3odFcioxEv7sSTxlpGCC9-My7cueHXV-d-  
JGvd3RVzUXrAftdTY3W_BLU4gc/exec"  
API_KEY = "nrt_8F2xQ9mL7vP3zK1cR6wT4yH0bN5sJ8"  
DASHBOARD_API_URL = "http://127.0.0.1:5000/api/nrt-status"  
GOOGLE_DRIVE_FOLDER_ID = "1och4pet9VHRFBu6DQJVRn5Oqy7LNUF_R"  
GOOGLE_DRIVE_FOLDER_URL = "https://drive.google.com/drive/folders/1och4pet9VHRFBu6DQJVRn5Oqy7LNUF_R"  
SMTP_SERVER = os.getenv("SMTP_SERVER", "mail.dtechdigital.com.my").strip()  
SMTP_PORT = int(os.getenv("SMTP_PORT", "465"))  
SMTP_USERNAME = os.getenv("SMTP_USERNAME", "support@dtechdigital.com.my").strip()  
SMTP_PASSWORD = os.getenv("SMTP_PASSWORD", "n^;J#O76;Xre")  
LOCAL_EMAIL_ALERTS_ENABLED = os.getenv("LOCAL_EMAIL_ALERTS_ENABLED", "0").strip().lower() in {"1", "true", "yes", "on"}  
ALERT_EMAIL_RECIPIENTS = [  
    address.strip()  
    for address in os.getenv("ALERT_EMAIL_RECIPIENTS", "").split(",")  
    if address.strip()  
]  
EMAIL_ALERT_THRESHOLD_HOURS = 24  
  
ALERT_THRESHOLD_HOURS = 48  
COLLECTION_INTERVAL_MINUTES = 40  
STARTUP_LAUNCHER_BASENAME = "NRT Monitor Collector"  
STARTUP_LAUNCHER_NAME = f"{STARTUP_LAUNCHER_BASENAME}.vbs"  
LEGACY_STARTUP_LAUNCHER_NAME = f"{STARTUP_LAUNCHER_BASENAME}.cmd"  
SINGLE_INSTANCE_MUTEX_NAME = "Global\\NRTMonitorCollectorSingleton"  
LOG_PATH = Path(__file__).with_name("agent.log")  
STATUS_PATH = Path(__file__).with_name("agent_status.json")  
STATUS_SUMMARY_PATH = Path(__file__).with_name("agent_status.txt")  
SITE_LOGS_DIR = Path(__file__).with_name("site_logs")  
  
CONNECTION_STRING = (  
    "DRIVER={ODBC Driver 17 for SQL Server};"  
    f"SERVER={SQL_SERVER};"  
    f"DATABASE={DATABASE};"  
    "UID=sa;"  
    "PWD=qwerty7890;"  
    "Trusted_Connection=yes;"  
)
```

5. **Run the app**

Place both program one in the Central Server, another in the client site

On the Central Server, run:

```
python app.py
```

On the client side, run:

```
python agent.py
```

Output:

127.0.0.1:5000/?site=test-site-nrt-v2-local-sqlexpress#site-detail

Sites below currently have failed NRT detection based on TB_TNG_CONTROL_BATCH.

Test Site

5 control batch rows

SITE DETAIL

Test Site (NRT-V2)

(local)\SQLEXPRESS | NRT-V2

NRT alert - 5 control batch records pending, oldest operdate is 120 hours old

TERMINAL COUNT

1

BATCH COUNT

5

LATEST OPERDATE

2026-05-07 15:29:46

OLDEST OPERDATE

2026-05-03 15:29:46

OLDEST AGE HOURS

120

RECEIVED AT

2026-05-08 15:48:00

Red

TB_TNG_CONTROL_BATCH

5 rows

Site ID	Terminal ID	Batch No.	Operdate	EOJ Date Time	Record Count	Age Hours	Status
D67	P01	105	2026-05-07 15:29:46	2026-05-08 15:29:46	20	24	
D67	P01	104	2026-05-06 15:29:46	2026-05-07 15:29:46	15	48	
D67	P01	103	2026-05-05 15:29:46	2026-05-06 15:29:46	8	72	
D67	P01	102	2026-05-04 15:29:46	2026-05-05 15:29:46	12	96	
D67	P01	101	2026-05-03 15:29:46	2026-05-04 15:29:46	10	120	